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Ciclo: 26-1
Materia: Terminología Especializada en Tecnología Inglés-Español
Curso: IT0627
Licenciatura: Interpretación y Traducción
Grupo: 6A

HW_10 Astrobiology, Agriculture, and AI; Vocab

(convert to 10 point scale)

Criteria	Excellent	Satisfactory	Needs Improvement
Task 1: Astrobiology and Ag	Response provided: a) Astrobiology as a discipline; b) the three current research areas; c) the Zoo hypothesis is correct?; d) Problems w/ existing ag?; e) crop rotation as a top priority; f) switch to a plant-based diet? Accurate, Complete, and Sufficient response	Mostly Accurate, Complete, and Sufficient response	Not Accurate, Complete, and/or Sufficient response
Task 2: AI	Response provided:: a) Analysis legit in the two papers?; plausible scenarios and methods?; c) opinion (2 paragraphs)	Mostly Accurate, Complete, and Sufficient response	Not Accurate, Complete, and/or Sufficient response
Task 3: Vocab 15 (English)	Complete definitions with Spanish or English translation	Partial definitions	Incomplete and missing definitions
Document Formatting and Presentation	Document is perfectly formatted according to Marron Style Guide .	Minor formatting errors present but overall follows Marron Style Guide.	Formatting does not follow the Marron Style Guide or is inconsistent throughout.

Assignment:

***HW_10 [Due: 09 Dec 2025]

<== due in two weeks; can use on the exam

--- Task 1

Answer the following questions related to three scientific journal articles. The articles can be found in your 3P folder under Class Materials.

**Focus on astrobiology

[01_Nature_Astrobiology.pdf]

1) What is the science of astrobiology?

2) Three important lines of research in astrobiology are i) defining a habitable environments, ii) identifying biosignatures, and iii) defining technosignatures. Briefly explain what is meant by each of these research areas

3) Read the Abstract of the article, "Is the apparent absence of extraterrestrial technological civilizations down to the zoo hypothesis or nothing? (<https://www.nature.com/articles/s41550-023-02134-2>). What are the authors suggesting?

**** Diversifying crop rotation increases food production, reduces net greenhouse gas emissions and improves soil health**

[02_Nature_Diversifying crop rotation.pdf]

1) What are the problems with conventional intensified food production systems

2) Why do the authors consider crop rotation as a "top priority" for agriculture? (i.e., what are the advantages of crop rotation?)

**** Associations between national plant-based vs animal-based protein supplies and age-specific mortality in human populations**

[03_Nature_plant-based vs animal-based protein supplies.pdf]

1) Based on the article, do you recommend a switch to a plant-based diet?

--- Task 2

Read/scan two reports on the future of artificial intelligence (AI). The first report, written by the AI Futures Project, is available online at <https://ai-2027.com/> and also as a .pdf (04_ai-2027.pdf). You should read/scan the online report. The second report, written by Richard David Hames, is available as a .pdf file ("05_The Future of AI - by Richard David Hames.pdf"). Compare and contrast the futures presented in these two white papers. Specifically,

1) What do think of the analysis presented in each paper?

2) Are the scenario(s) presented plausible?

3) Are the authors and their methods credible?

Write an opinion piece (~2+ paragraphs) expressing your thoughts. Your opinion piece should be written in English.

--- Task 3

Define the following terms in English

perceptron

artificial neural network

Large Language Model (LLM)

supervised learning

- unsupervised learning
- reinforcement learning with human feedback (RLHF)
- emergence
- Application Programmer Interface (API)
- Software Development Kit (SDK)
- Python
- token (as used in AI)
- prompt engineering
- machine translation with post editing (MTPE)
- bilingual evaluation understudy (BLEU)
- metric for evaluation of translation with explicit ordering (METEOR)

NB. Marron's modified APA format applies (see Class Materials in Teams). As always, assignments must be printed (i.e., hard copy) and submitted to the instructor on the due date (unless duly noted). Students should save copies of all assignments in PDF format.